

What is biochip 2

- There are different names for the microarrays, like DNA/RNA Chips, BioChips or GeneChips.
- The array can be defined as an ordered collection of microspots, each spot containing a single defined species of a nucleic acid.
- The microarray technique is based on hybridisation of nucleic acids.
- There exist two variants of the chips:

cDNA microarrays

oligonucleotide arrays

Properties of ceramics and glass 3

- Ceramic materials by definition are ionically bound; they are hard materials, both electrically and thermally very stable.
- Included in this category are, for example, Al_2O_3 , Si_3N_4 , MgO , SiO_2 , Na_2O , CaO and ZrO_2 .
- Ceramics are characterised by being hard yet brittle, therefore, in many cases, they are used as composites where they are mixed with other materials (e.g. metals) to increase their mechanical performance.

Properties:

- High optical transparencies.
- Silica aerogels are made of pores of about 10 nm arranged in distances between 10 and 100 nm.
- They are resistant and chemically inert to liquid metals, heat resistant up to 1 200 °C and non-toxic.
- Biomedical applications, such as substrates for cell growth and analysis.

Features of sensors 3

- A sensor is a device capable of recognizing a specific chemical species and signaling the presence, activity or concentration of that species in solution through some chemical change.

Feature of sensor:

- **Biosensor = bioreceptor + transducer.**
- **A device that utilizes biological components e.g. enzymes to indicate the amount of a biomaterial.**
- **A biosensor is an analytical device, used for the detection of an analyte, that combines a biological component with a physicochemical detector.**
- **The bioreceptor is a biomolecule that recognizes the target analyte whereas the transducer converts the recognition event into a measurable signal.**
- **The sensitive biological element**

(e.g. tissue, microorganisms, organelles, cell receptors, enzymes, antibodies, nucleic acids, etc.)

is a biologically derived material or biomimetic component that interacts (binds or recognizes) with the analyte under study.

Application of biosensor in environment 3

With high specificity and sensitivity, biosensors provide an exceptional analytical system for the monitoring of environmental pollutants.

They are inexpensive and attractive alternative to the conventional analytical techniques, which are capable of providing real-time online monitoring.

With the diverse effect of these pollutants on the biological system, a number of biosensors have been developed and are still in progress.

Among toxic compounds, determination of heavy metals, phenolic compounds, mercury, organophosphorus, and carbamate pesticides is the major concern, considering their extensive contribution in increased pollutant level.

Biosensor or biodevices can be used as environmental quality monitoring tools in the assessment of physical, chemical and biological monitoring of pollutants.

The major applications of biosensors are monitoring of various pollutants including heavy metals, organic and inorganic pollutants, toxins, antibiotics and contaminating microorganisms.

Heavy metals

Heavy metals, i.e. copper (Cu), cadmium (Cd), mercury (Hg), lead (Pb), zinc (Zn), etc., are recently the major cause of serious environmental pollution problems.

These are known for their high toxicity and bioaccumulation attribute in the food chain.

A number of bacterial biosensors have been developed for the determination of heavy metals in different environmental samples using enzyme and DNA as bioreceptor and optical and electrochemical as transduction system.

Biochemical oxygen demand (BOD)

Biochemical oxygen demand is one of the significant parameters used in the estimation of the concentration of biodegradable organic pollutants present in a water sample

Nitrogen compounds

Nitrogen compounds, i.e. nitrates, are widely used by food manufacturing industries as preservatives (increase shelf-life) and chemical fertilizer industries as fertilizer (increase the fertility of the soil).

Polychlorinated biphenyls (PCBs)

PCBs are toxic organic compounds, universal environment pollutants; hence several countries banned their production long time ago.

Such compounds are highly lipophilic nature, therefore abundant chances of accumulating in the food chain.

Phenolic compounds

Phenols and their derivatives, i.e. chlorophenol are distributed commonly in the environment.

Such compounds are mainly used in the production of dyes, drugs, plastics, pesticides, detergents, etc..



Detail on bioreceptors 5

Biosensors:

The sensor which shows the reaction in efficiency is called biosensor. Biosensor can be enzyme. In other words enzymes can act as a biosensor. It is a device which makes the detection of analyte which connects to a biological part By a detector whose is physiochemical. The sensitive elements can be made by biological engineering. This kind of detector elements are very useful and known as biosensors. Biosensor reader are the devices which are electronically joint with biosensor.

Biosensor system:

Biosensors are consisted of two site one is recognition site which is also called bioreceptors. Recognition component is use the biomolecules for the interaction with analyte which is interested for the wok. Main purpose of biosensor is to make efficient suitable care.

Biosensor types:

- Electronic biosensors
- Pyroelectric biosensor
- Gravimetric biosensor
- Optical biosensors
- Electrochemical

Uses:**To quantify the glucose focus, three unique transducers can be utilized:**

- An oxygen sensor that measures oxygen fixation
- A pH sensor that measures the corrosive (gluconic corrosive) creation
- A peroxide sensor that measures H₂O₂ fixation.
- An oxygen sensor is a transducer that believers oxygen fixation into electrical current.
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Cholesterol monitoring:

1. Low motivating force to utilize
2. Cost high
3. What do you do with the data?
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Contemplations in Biosensor Development

Once an objective analyte has been recognized, the real assignments in building up a biosensor includes:

- Selection of an appropriate bioreceptor atom
- Selection of an appropriate immobilization strategy
- Selection of an appropriate transducer
- Designing of biosensor considering estimation range, linearity, and minimization of obstruction
- Packaging of biosensor

Applications of biosensors:

Biosensors are gadgets including a natural component and a physiochemical indicator that are utilized to distinguish analytes. These instruments have an extensive variety of utilizations running from clinical through to natural and farming. The gadgets are likewise utilized as a part of the sustenance business. A few cases of the fields that utilization biosensor innovation include:

- General medicinal services observing
- Screening for ailment
- Clinical investigation and analysis of sickness
- Veterinary and agrarian applications
- Modern preparing and checking
- Ecological contamination control
- In sustenance preparing, checking, nourishment genuineness, quality and security
- Quality and security, support of nourishment items and handling

Nourishment verification and observing with objective and steady estimation of sustenance items, in a financially savvy way, are attractive for the nourishment business.

Therefore improvement of biosensors in light of the interest for basic, ongoing, particular and reasonable procedures is apparently auspicious. Observing of maturing of lager utilizing enzymatic biosensors, in light of cobalt phthalocyanine – amid capacity .Biosensors are utilized for the location of pathogens in sustenance.

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In maturation forms :

In maturation businesses, process wellbeing and item quality are essential.

Along these lines viable checking of the maturation procedure is basic to create, upgrade and keep up natural reactors at greatest viability. Biosensors can be used to screen the nearness of items, biomass, protein, counter acting agent or side-effects of the procedure to in a roundabout way measure the procedure conditions. Biosensors unequivocally control the maturation business and deliver reproducible outcomes because of their basic instrumentation, impressive selectivity, low costs and simple robotization. Biosensing innovation for manageable sustenance wellbeing The term nourishment quality alludes to the appearance, taste, smell, wholesome esteem, freshness, flavor, surface and synthetic concoctions.

Savvy observing of supplements and quick screening of organic and synthetic contaminants are of principal significance, with regards to nourishment quality and security. Glutamine is the quick and dirty of essential capacities, for example. Patients insufficient in glutamine experience

the ill effects of pathologies, for example, malabsorptive clutters and must be supplemented, to enhance insusceptible capacities. Glutaminase-based microfluidic biosensor chip with a stream infusion examination for electrochemical recognition has been utilized for location in maturation process.

In restorative field:

In the teach of restorative science, the utilizations of biosensors are developing quickly. Glucose biosensors are generally utilized as a part of clinical applications for analysis of diabetes mellitus, which requires exact control over blood-glucose levels. Blood-glucose biosensors use at home records for 85% of the colossal world market. Biosensors are being utilized unavoidably in the restorative field to analyze irresistible sicknesses. A promising biosensor innovation for urinary tract disease (UTI) conclusion alongside pathogen distinguishing proof and against microbial powerlessness is under examination. A novel biosensor, in light of hafnium oxide (HfO₂), has been utilized for beginning time discovery of human interleukin (IL).

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IN food industry:

- Food processing industry faces various challenges; one of the foremost challenges is the need for quick and cost effective methods to detect the presence of allergenic components and pathogens in the food.
- Biosensors pave way for the rapid detection of pathogens, allergens as well as pesticide residues in food.
- Detection of contaminants, verification of product contents, product freshness and monitoring of raw materials conversion are the areas of potential biosensor applications.
- Biosensors have the potential to produce an analytical revolution to resolve the challenges in the agricultural and the food industries.

- The determination of chemical and biological contaminants in foods is of paramount importance to the health of food because, unlike the contamination of a physical nature, they cannot be displayed.
- Development of biosensors to the analysis of the quality of food, since they have proven to be an extremely viable alternative to traditional analytical techniques such as chromatography.
- In the area of food the interest in the development of biosensors mainly focuses on analysis of food security (detection of compound contaminants, allergens, toxins, pathogens, and additives etc.) Food composition and online process control.

Biosensors for diabetes applications

Glucose as diabetes biomarker

- About 3% of the population worldwide suffers from diabetes, a leading cause of death, and its incidence is growing fast.
- Diabetes is a syndrome of disordered metabolism resulting in abnormally high blood sugar levels.
- Optimal management of diabetes involves patients measuring and recording their own blood glucose levels.
- Under normal physiological condition, the concentration of fasting plasma glucose is in the range 6.1–6.9 mmolL⁻¹, so the variation of the blood glucose level can indicate diabetes mellitus, besides other conditions.
- Association recommends that insulin-dependent type 1 diabetics self monitor blood glucose 3–4 times daily, while insulin-dependent type 2 diabetics monitor once-daily.
- Therefore it is necessary to develop a simple, sensitive, accurate, micro-volume and
- low-cost approach for glucose analysis which is appropriate for rapid field tests and is also effective as an alternative to the existing methods.

First-generation of glucose biosensors (Clark enzyme electrode)

- First proposed in 1962 by Clark and Lyons
- These sensors were initially based on an electrochemical approach, which used the enzyme glucose oxidase (GOx).

- Electrochemical sensors were chosen for blood-glucose measurements due to their high sensitivity, on the order of μM to mM , good reproducibility and ease of fabrication at relatively low cost.
- GOx catalyses the oxidation of glucose to gluconolactone in the presence of oxygen, while producing hydrogen peroxide (H_2O_2) and water as by-products.
- Measurements of peroxide formation have the advantage of being simpler, especially when miniature devices are being considered.
- The first commercially successful glucose biosensor using Clark's technology was the Yellow Springs Instrument Company analyzer (Model 23A YSI analyzer) for the direct measurement of glucose in 1975, which was based on the amperometric detection of hydrogen peroxide.
- This analyzer was almost exclusively used in clinical laboratories because of its high cost due to the expensive platinum electrode.

Explain self adhesive and why diatoms are good model for biomimetic membrane 5

Self-healing adhesives

- Diatoms are a type of algae with nanostructured amorphous silica surfaces.
- Diatoms are unicellular algae that have the capability to colonize any natural and manmade submerged surfaces.
- Diatoms are a large group of eukaryotic microalgae that have a rigid cell wall made of silica (SiO_2).
- Diatoms are often among the first organisms to colonize underwater surfaces, because their adhesive mucilage enables the cells to attach to substrata of a very wide range of chemical composition (organic, inorganic, natural or man-made)./,,,W
- Surface adhesion of diatoms belonging to the raphid pennate type requires the secretion of so-called adhesive mucilage strands through a dedicated slit in the silica cell wall, which is termed the raphe.

Self-healing adhesives

- Adhering cells can then move across the surface, which is accomplished through an actinmyosin dependent cytoskeletal motor that translocates the adhesive mucilage strands in a rearward direction through the raphe, thereby moving the cell forward.
- Some diatom species have evolved strong self-healing underwater adhesives. Some are

free-floating, others have adhesive properties in water: for instance, diatoms in the Antarctic seas can attach to ice.

- Others secrete viscous mucilage which binds colonies together while protecting the silica shells from wear as they rub against each other.
- Molluscs are another species that have adhesive properties under water.
- Both diatoms and molluscs have strong underwater glues that can also resist stress and self-heal.

They serve as a biomimetic model for self-healing materials.

Self-healing adhesives

- These natural adhesives and their self-healing properties are due to the properties of the proteins contained in them.
- These proteins have sacrificial bonds that allow the molecule to be reversibly stretched by breaking and re-bonding.
- This sacrificial bond behaviour has been observed in many other materials, such as wool. Closing the wounds with sticky materials has been introduced to prevent bleeding and induce the wound healing process.
- Many types of surgical adhesives including tissue adhesives have been developed to be a suitable alternative for sutures and staples.
- A new approach which aims at producing bio-adhesives by mimicking the nature along with applying nanotechnology methods.

Self-healing adhesives

- Engineering stable underwater adhesives currently poses a major technical challenge, because most man-made adhesives fail in wet conditions, owing to chemical modification of the adhesive or its substrate.
- Diatoms produce adhesives that are extremely strong and robust, in both fresh- and seawater environments.
- AFM (Atomic Force Microscopy) study of living diatoms, phase imaging and force spectroscopy experiments revealed characteristics of these natural adhesives that may be of use in designing man-made adhesive analogues that function in wet environments.
- The AFM allows for investigations of micromechanical properties of the cell surface, for example viscoelastic properties, adhesion forces and hardness measurements, in physiological conditions.
- Experiments with three diatom species that produce outstanding natural

adhesives: *Eunotia sudetica*, *Navicula seminulum* and an unidentified species

What is spectroscopy and write X-ray method in detail 5

Spectroscopy is defined as the branch of science that is concerned with the investigation and measurement of spectra produced when matter interacts with or emits electromagnetic (EM) radiation.

Spectroscopy is a technique that uses the interaction of energy with a sample to perform an analysis.

Depending on the wavelength of the electromagnetic used and the type of interaction with matter that occurs (absorption, scattering, etc.), different spectra are measured from which much information can be inferred

X ray:

X-rays were discovered in 1895 by German physicist Roentgen.

X-ray methods involve exciting a sample either with X-rays (creating more X-rays) or with electrons (creating X-rays).

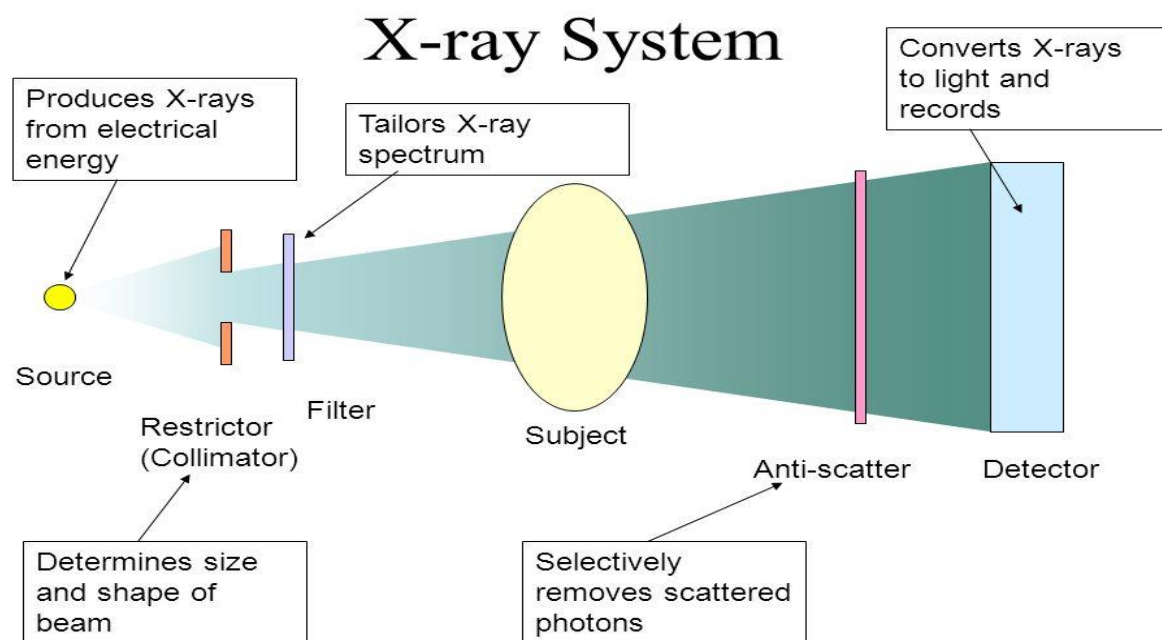
X-rays can be also generated by bombarding a sample with alpha particles.

The energy of emitted X-rays is equal to the difference between the binding energies of the electrons involved in the transition.

There are various methods that use X-rays: X-ray fluorescence (XRF), X-ray diffraction (XRD), etc.

In the context of nanomaterials, the most important method is small-angle X-ray scattering (SAXS) analysis.

Like XRD, this method is based on the principle of scattering of X-rays.



Spanning a wide range of research fields, such as biology, chemistry, physics, and engineering, nanotechnology is uniquely interdisciplinary in character.

The common factor in nanotechnology is the lateral dimension of the structures studied.

Defined as materials being in, or having components in, the one billionth (10^{-9}) of a meter range, nanotechnology research and development relies on accurate measurement of atomic and molecular distances within structures ranging from semiconductor devices to nano-powders.

The dimensions of X-ray wavelengths are of the same magnitude as the size of nanostructures, X-ray diffraction (XRD) and associated techniques are primary tools for the nanotechnology researcher.

X-ray reflectometry (XRR) determines layer thickness, roughness, and density.

High-resolution X-ray diffraction can measure layer thickness, roughness, chemical composition, lattice spacing, relaxation and more.

X-ray diffuse scattering is used to determine lateral and transversal correlations, distortions, density, and porosity.

A new advance in X-ray imaging has revealed the dramatic three-dimensional shape of gold nanocrystals, and is likely to shine a light on the structure other nano-scale materials.

The new technique improves the quality of nanomaterial images, made using X-ray diffraction, by accurately correcting distortions in the X-ray light.

Most nanomaterial imaging has been done using electron microscopy. X-ray imaging is an attractive alternative as X-rays penetrate further into the material than electrons and can be used in ambient or controlled environments.

Detail on application of biosensors 10

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BT405

Quantum Continent

Role of Nanotechnology in Diagnostics:

Several nanoparticles have been used for diagnostics. Of these, the most frequently used are gold nanoparticles, QDs, and magnetic nanoparticles.

Nanodiagnostics is the new term that describes use of methods and techniques of nanotechnology and its principles for diagnostics purposes.

It includes, although not limited to, the manipulation and assessment of single molecule, size reduction of systems and platforms to make use of nanoscale properties obtainable from interactions between surfaces and biomolecules.

Nanodiagnostics is an evolving application of nanoscale technology to meet the demand of clinical diagnostics, determining disease state, any predisposition to such, the pathology of the condition and the identification of causative organisms

Five uses of nanotechnology in daily life

Food & packaging

Textile and clothing

Sports & leisure, household

Computing & electronics

Transport, building, construction

Medical

Environment The total market for nanobiotechnology products was \$19.3 billion in 2010 and is growing at a compound annual growth rate (CAGR) of 9% to reach a forecast market size of \$29.7 billion by 2015.

Medical applications, including drug delivery and microbicides, dominate today's market, with sales of \$19.1 billion in 2010. This market segment is growing at a compound annual growth rate (CAGR) of 8.7%, and is forecast to reach sales of \$29 billion by 2015 (BCC).

More and more consumer products are branded worldwide as nano.

The most common nanoscale substance used in consumer products is nano-silver (anti-bacterial and anti-fungal properties)

Baby bottle nipples, door handles, food contact products, interiors of refrigerators and washing machines, clothing, bedding, paints, cleaning products, food storage containers, bandages, medical devices, mattresses, and children's teddy bears

Spectroscopy and X-ray

siRNA Drug Delivery System

Among these clinical trials, most siRNAs were administered by local delivery, typically via the [intravitreal](#) or [intranasal routes](#). However, local delivery may not be appropriate for all diseases. Under some circumstances, systemic drug administration by [intravenous](#) Compared to chemotherapeutic anti-cancer drugs, there are a lot of advantages of siRNA drug. Due to the special mechanism of siRNA, it has four advantages as a potential cancer therapeutic strategy. The first is its high degree of safety. siRNA acts on the post-translational stage of gene

expression, so it does not interact with DNA and thereby avoids the mutation and teratogenicity risks of gene therapy. The second advantage of siRNA is its high efficacy. In a single cancer cell, siRNA can cause dramatic suppression of gene expression with just several copies.

Types of Glucose Biosensors

Types of Glucose Biosensors

Enzymatic glucose biosensors

First generation glucose biosensor

Second

Non-enzymatic glucose biosensors

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Second-generation of Glucose Biosensors (Mediated Biosensors)

Limitations of the first generation glucose biosensors were overcome by using mediated glucose biosensors, second generation glucose sensors.

The improvements were achieved by replacing oxygen with non-physiological electron acceptors, called redox mediators that were able to carry electrons from the enzyme to the surface of the working electrode.

Third-generation of Glucose Biosensors

This feature allows a 3rd generation glucose sensing mechanism, which is independent of mediators or oxygen.

The third-generation glucose biosensors are reagentless and based on direct transfer between the enzyme and the electrode without mediators.

Instead of mediators with high toxicity, the electrode can perform direct electron transfers using organic conducting materials based on charge-transfer complexes.

Optical Microscope and its resolution power

- An optical microscope uses visible light (i.e. electromagnetic radiation) and a system of lenses to magnify images of small samples.
- For this reason, it is also called a light microscope. Optical microscopes are the oldest and simplest of the microscopes.
- The resolution limit of an optical microscope is governed by the wavelength of visible light.
- Visible light is the part of the electromagnetic spectrum with wavelengths between 400 and 700 nm.
- The resolving power of an optical microscope is around 0.2 μm or 200 nm: thus, for two objects to be distinguishable, they need to be separated by at least 200 nm.

Single objects smaller than this limit are not distinguishable: they are seen as fuzzy objects. This is known as the diffraction limit of visible light.

Microarray

Application of Biosensors

- Other microscopes have been designed which use other beams: rather than light, they use electron beams to illuminate the sample.

- Electron microscopes have much greater resolving power than light microscopes that use electromagnetic radiation and can obtain much higher magnifications of up to two million times, while the best light microscopes are limited to magnifications of 2 000 times.
- There are various types of electron microscopes, such as the scanning electron microscope (SEM) and or the transmission electron microscope (TEM).

Scanning Tunnelling Microscope (STM) can create detailed 3D images of a sample with atomic resolution.